

LORDS
Institute Of Engineering and Technology
NBA Accredited | NAAC 'A' Grade | NABL Accredited
Estd. 2003
UGC AUTONOMOUS

REVIEW 1

Predictive Healthcare Risk Stratification Using Machine Learning

Student Names :
Mohammed Azaan Ahmed
Arham Ahmed khan
Abdullah Mohammed

Roll Numbers :
160923733116
160923733075
160923733074

Department:CSE - B
Guide Name: Mr. Parvez Moin

01

Abstract

This project builds a predictive analytics system that stratifies anonymized patient populations into risk groups using clinical, demographic, utilization, and longitudinal health features. Several supervised learning algorithms are compared, with explainability methods used to identify influential variables. The platform presents risk probabilities and cohort-level analytics rather than autonomous clinical decisions. Evaluation covers discrimination, calibration, subgroup performance, and appropriate handling of missing and imbalanced data.

In simple words: the system learns from past anonymized patient data, groups patients by risk, and shows why.

02

Objective

Main objective: to develop a machine learning based system that uses anonymized healthcare data to predict and classify patients or patient groups into different levels of risk.

- Use clinical, demographic and health history data
- Train different machine learning models for risk prediction
- Compare the performance of the models
- Identify which factors influence the prediction
- Handle missing and unbalanced data properly
- Check whether predictions are reliable across different patient groups
- Present risk probabilities and risk categories clearly

Literature Survey: Machine Learning in Healthcare Risk Prediction

Author / Year	Method Used	What the Study Did	Main Finding
Chen, Sun, Dittus et al., 2016 IEEE J. Biomed. Health Inform.	Patient stratification using electronic health records	Grouped patients in a chronic disease management program using health records.	Health records were used to group patients.
Beaulieu-Jones et al., 2021 npj Digital Medicine	Deep learning on administrative hospital data	Predicted in-hospital death, long stay and 30-day readmission from 42.9 million admissions.	AUC 0.89, 0.82 and 0.71. Models should add new insight, not copy clinicians.
EHR-based ML model for gastrointestinal bleeding Gastroenterology	Machine learning versus standard clinical risk scores	Compared the model with Glasgow-Blatchford and Oakland scores.	AUC 0.92 versus 0.89 for the standard scores.

Machine learning can identify patterns and predict patient risk. Model performance depends on the type and quality of data.

04 · LITERATURE

Literature Survey: Explainable and Reliable Healthcare AI

Author / Year	Method / Focus	Main Finding	Relevance to Our Project
SHAP-based explanations for chronic kidney disease risk prediction ACM conference 2025	Random Forest and XGBoost with SHAP and calibration	SHAP showed which test values mattered most; results were consistent across sex and age.	Explanation, calibration and group checks
ShapleyVIC PLOS Digital Health 2024	Variable importance for fair risk prediction	More stable importance results than random forest or XGBoost on small samples; helped justify leaving out race.	Finding important factors fairly
Anderson and Visweswaran, 2025 arXiv	Effect of enforcing fairness on SHAP explanations	Improving fairness across racial groups can change which factors rank as important.	Check accuracy, fairness and explanations together

Existing systems often focus mainly on prediction accuracy. Our project also considers why a prediction is made, probability reliability, and performance across different groups.

05

Existing Systems / Approaches

Useful solutions already exist.

01 Rule Based Risk Scoring

Uses predefined medical rules to assign risk scores.

02 Statistical Models

Methods such as Logistic Regression estimate patient risk.

03 Machine Learning Models

Random Forest and Gradient Boosting learn patterns from historical data.

04 Healthcare Decision Support Systems

Show risk information to healthcare professionals so they can decide next steps.

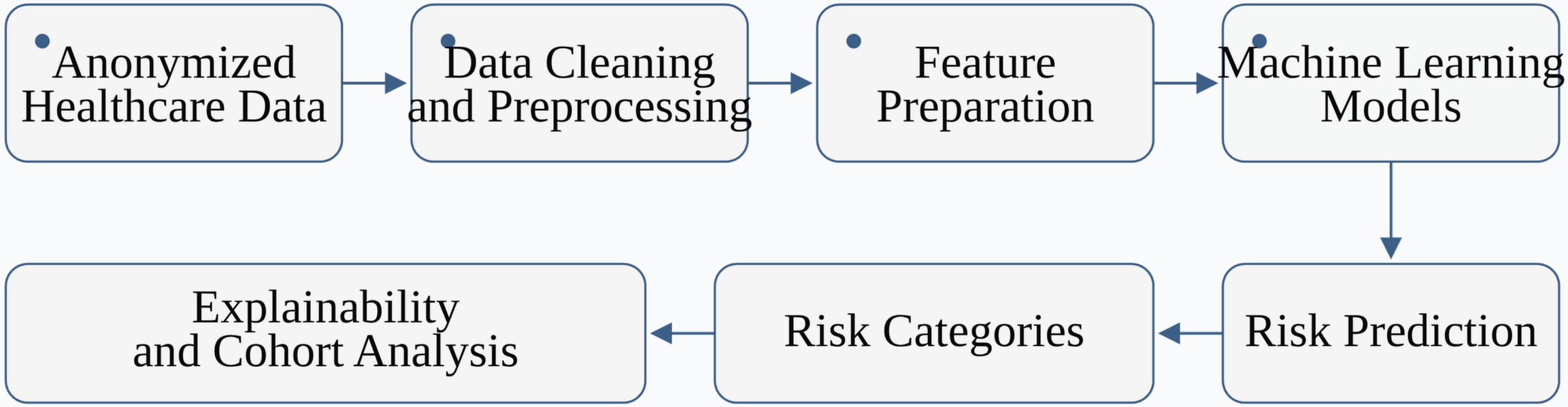
06 · COMPARISON

Existing Approaches vs Proposed Approach

Feature	Existing Approaches	Proposed Approach
Risk prediction	Often included	Planned
Multiple model comparison	Not always included	Planned
Important factors behind predictions	Not always explained	Planned
Missing data handling	Not always explicit	Planned
Imbalanced data handling	Not always explicit	Planned
Reliability of risk probabilities	Not always checked	Planned
Performance across different groups	Not always checked	Planned
Cohort level analysis	Not always included	Planned

07 · ARCHITECTURE

Proposed Design / System



Low Risk | Moderate Risk | High Risk

The system provides risk predictions and analytical insights. It does not diagnose patients or make autonomous medical decisions.

Limitations

- Data Dependency: Results depend on the quality and availability of healthcare data.
- Data Imbalance: Unequal data across risk groups may affect predictions.
- Generalization: Performance may vary across different patient populations.
- Clinical Validation: Further testing is required before real world use.
- Decision Support: The system cannot replace professional medical judgment.

Advantages

- Risk Stratification: Classifies patients into Low, Moderate and High Risk groups.
- Multiple Model Comparison: Compares different machine learning algorithms for risk prediction.
- Explainable Predictions: Identifies the factors influencing the model’s predictions.
- Reliable Evaluation: Evaluates prediction performance, calibration and subgroup performance.
- Cohort Level Insights: Provides risk probabilities and analytical insights for patient populations.

09 · TOOLS

Technologies and Algorithms

Technologies

Python, Pandas, NumPy, Scikit-learn
Matplotlib / Seaborn
Jupyter Notebook / VS Code

Machine Learning Algorithms

Logistic Regression, Decision Tree
Random Forest

Explainability

SHAP shows how much each factor pushed the prediction.
Feature Importance

Evaluation

Accuracy, Precision, Recall, F1 Score, ROC-AUC, Confusion Matrix
Calibration checks how reliable the predicted probability is.

Thank You!